

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** building high-scale backend systems across Google, YouTube, and Whatnot. Owned database infrastructure and observability products from design through production for platforms serving billions of users. Led teams of 10+ engineers and drove measurable improvements in system reliability and developer productivity. Background in distributed databases using **Vitess**, **MySQL**, and **PostgreSQL**. Adept at instrumenting systems for observability and creating customer-facing insights products. Seeks to apply deep backend and observability expertise to database tooling challenges. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Go, Python, Ruby, C++, C#, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, Docker, GCP, Pub/Sub, Bigtable

Databases: Vitess, MySQL, PostgreSQL, Spanner, Vitess

Observability: OpenTelemetry, Prometheus, Grafana, ELK Stack, Distributed Tracing

Platforms: Ruby on Rails, REST APIs, gRPC, Kafka, Redis

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Leading backend infrastructure development for video platform serving billions of daily users, focusing on performance-critical services and data pipelines.
- Applying observability and database expertise to improve reliability of core video delivery and recommendation systems at massive scale.
- Building internal tooling and frameworks to enhance developer productivity across distributed engineering teams.
- Evaluating trade-offs between distributed database technologies for specific workload patterns and consistency requirements.
- Designing APIs and integrations for new platform capabilities supporting content creator workflows.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Architected Ruby on Rails platform serving live marketplace where hundreds of thousands of users buy and sell across simultaneous streams.
- Shipped developer experience tools including interactive debugging interfaces that cut mean time to diagnosis from hours to minutes.
- Built comprehensive observability infrastructure, instrumenting **Vitess** and **PostgreSQL** clusters to emit performance telemetry for internal dashboards.
- Developed unified logging pipeline aggregating application metrics, query performance data, and infrastructure health across **Kubernetes** clusters.
- Led platform reliability initiative reducing production incidents by 60% through systematic observability improvements and on-call tooling.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 10 engineers building search ranking infrastructure serving google.com and contributing to quality improvements on major ranking launches.
- Designed ML feature pipeline processing petabyte-scale training data, reducing model staleness by 30% and improving ranking freshness.
- Established distributed tracing standards and observability practices adopted across three infrastructure teams, cutting incident resolution time significantly.

- Mentored engineers on distributed systems design and observability tooling, promoting four senior engineers during tenure.
- Partnered with cross-functional teams to define data infrastructure requirements for large-scale content indexing and retrieval.

Software Engineer | Google | 02/2011 - 02/2015

- Implemented ranking features for web search handling millions of queries per second, contributing to measurable quality improvements in A/B experiments.
- Optimized C++ code paths in serving infrastructure, reducing p99 latency by 15% for top traffic patterns during peak periods.
- Built automated testing frameworks and canary analysis tools preventing production issues from reaching majority of users.
- Developed internal dashboards surfacing search quality metrics and infrastructure health to on-call engineers.
- Shipped three major ranking improvements with measurable user engagement gains in controlled experiments.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Developed Windows client features using **C#** and **.NET Framework** for enterprise productivity applications with millions of users.
- Collaborated with program managers to define technical requirements and deliver features on schedule for release milestones.
- Conducted performance analysis and memory optimization for large-scale data processing components, achieving significant throughput gains.
- Authored technical documentation and internal tools improving team code review efficiency.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Contributed to feature development for Windows desktop application, implementing UI components and unit tests.
- Participated in code reviews and agile ceremonies while learning enterprise software development practices.

Education

B.S. Honors

Virginia Tech | 2004 - 2007